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CompareService

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OCRService

AI model & OCR Services Documentation

CompareService

Overview

Description
Service class for string comparison. Implements ICompareService. Provides Levenshtein distance and similarity percentage calculations.

Methods

LevenshteinDistance

Signature	Description
public async Task<int> LevenshteinDistance(string s, string t)	Calculates the Levenshtein distance between two strings (minimum number of insertions, deletions, or substitutions).

SimilarityPercentage

Signature	Description
public async Task<double> SimilarityPercentage(string? s, string? t)	Returns a percentage (0-100) representing similarity between two strings based on Levenshtein distance. Returns 0 if either string is null.

MLModel AI Controller Module

Overview

Description
Module for creating image classification ML models with training and error troubleshooting tips.

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Videos & Resources

Resource	Link / Notes
YouTube Tutorial	https://www.youtube.com/watch?v=fxyIefV8cuw
MLNet Meta File	Download and place at: C:\Users\hp\AppData\Local\Temp\MLNET\resnet_v2_50_299.meta Download Link

Notes (English / Arabic)

Language	Instructions
English	When training Image Classification, if it breaks, check error screen below. Likely manual download of meta file needed. Place in correct path, reopen project, train again.
Arabic	لما يعلق في التدريب اقرأ شاشة الايرور، في الغالب ستحتاج Image Classification عند عمل موديل صور تحميل الملف يدويا ووضعه في المسار الصحيح، ثم أغلق وافتح المشروع وأعد التدريب.

OCR Project

Module	Libraries / Notes
Image OCR	Extract text from images using SixLabors.ImageSharp and Tesseract. Requires downloading ara.traineddata and eng.traineddata. Place in `tessdata` folder inside project.
PDF OCR	Extract text from PDFs using free .NET libraries (supports English & Arabic).

OCRService

Overview

Description
Service for extracting text from images using Tesseract OCR and ImageSharp. Supports preprocessing, grayscale conversion, and custom cropping for ID Cards.

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Namespaces Used

Namespace
Tesseract
SixLabors.ImageSharp
SixLabors.ImageSharp.Processing
Microsoft.AspNetCore.Http

Core Methods

Method	Description
ReadAllTextAsync(IFormFile imageFile)	Reads all text from uploaded image without cropping.
ReadCropped_MarginXY_TextAsync(string grayPath, float marginXRatio, float marginYRatio)	Crops image using margin percentages, then extracts text.
ReadCropped_Custom4_TextAsync(string grayPath, float xL, float xR, float yU, float yD, string? lang = null)	Crops by custom ratios (Left, Right, Top, Bottom) and applies OCR.
ReadGrayTextAsync(IFormFile imageFile)	Converts uploaded image to grayscale and returns saved path.

Shortcut Methods for ID Cards

Method	Description
ExtractTextFrom_Cropped_IDNumberAndBirth_Async(string grayPath)	Extracts ID Number & Birth Date
ExtractTextFrom_Cropped_ExpiryDate_Async(string grayPath)	Extracts Expiry Date
ExtractTextFrom_Cropped_FullEnName_Async(string grayPath)	Extracts Full English Name
ExtractTextFrom_Cropped_FullArName_Async(string grayPath)	Extracts Full Arabic Name
ExtractTextFrom_Cropped_Gender_Async(string grayPath)	Extracts Gender
ExtractTextFrom_Cropped_Nationality_Async(string grayPath)	Extracts Nationality
ExtractAllTextDataFrom_IDCardGray_Async(string grayPath)	Returns all extracted ID Card data as DTO
DeleteTempFile1(string path)	Deletes temporary image file